Blockchain

ASSIGNMENT – 1

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BATCH-27

QUESTION1:Objective:

Tolearnblockchaininteractionbycreatingacryptocurrency

wallet,checkingwalletbalance,andsimulatingtransactionsusing

PythonandWeb3
Requirements:•InstallPython3.x

- SetupVSCodewithPythonextension
- InstallrequiredPythonlibraries:
- pipinstallweb3
- Useatestblockchainnetwork(EthereumSepolia/

Ganache local blockchain)

- Basicunderstandingofblockchainwalletsandprivatekeys

PracticalDescription:

Step1:EnvironmentSetup

- InstallPythonandVSCode
- InstallWeb3.pylibrary
- CreateaPythonfilenamedwallet_interaction.py

Step2:WalletandBlockchainInteractionScript
CreateaPythonscriptthat:•Connectstoablockchainnetwork

• Loadsawalletusingaprivatekey

• Fetcheswalletaddress

• Checkswalletbalance

• Demonstratestransactionpreparation(withoutrealfunds)

Code:

```
importtkinterastk
my_balance=10.0
x_balance=2.0
root=tk.Tk()
root.title("WalletSimulation")
root.geometry("400x300")

defupdate_ui():
  my_label.config(text=f"{my_balance:.2f}ETH")
  x_label.config(text=f"{x_balance:.2f}ETH")

defsend_money():
  globalmy_balance,x_balance
  amount_text=entry.get()

  ifamount_text=="": return

  amount=float(amount_text)
```

```
  ifamount<=my_balance:
    my_balance-=amount
    x_balance += amount
    update_ui()
    entry.delete(0,tk.END)

# - - - UI -----
tk.Label(root,text="MyWalletBalance").pack()
my_label=tk.Label(
  root,
  text="0ETH",
  font=("Arial",16),
  relief="solid",
  width=20,
  height=2
)
my_label.pack(pady=5)

tk.Label(root, text="X Wallet Balance").pack()
x_label=tk.Label(
  root,
  text="0ETH",
  font=("Arial",16),
  relief="solid",
  width=20,
  height=2
)
```

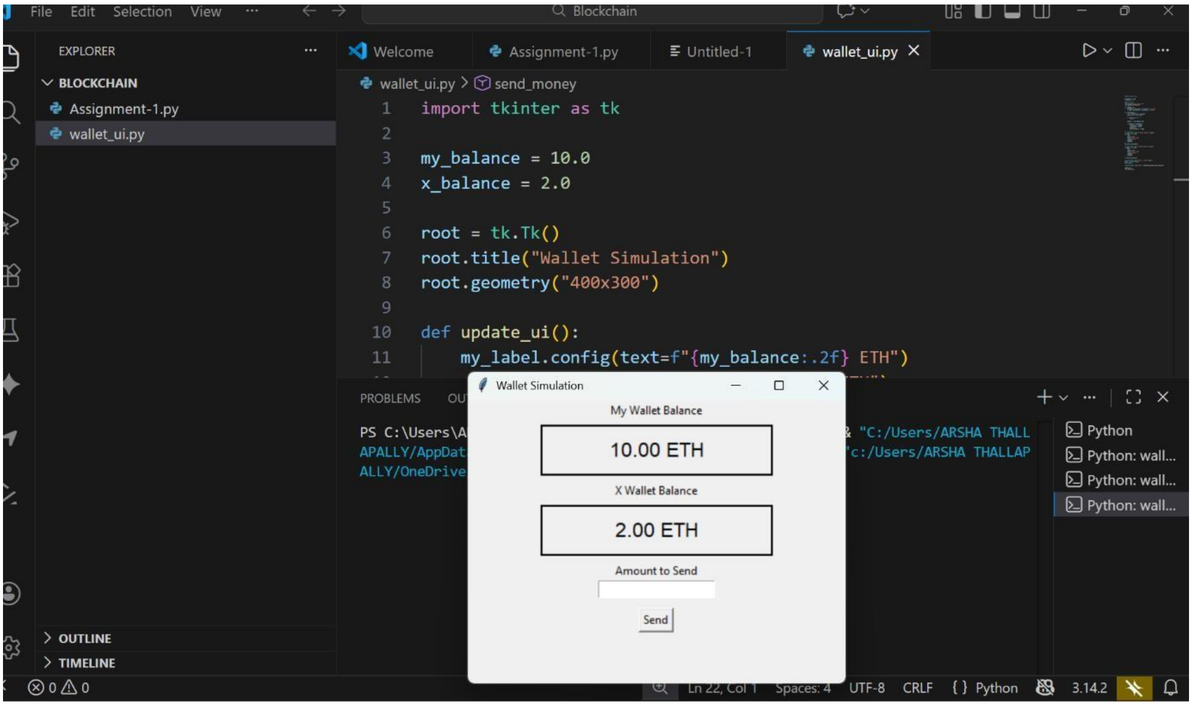
```
x_label.pack(pady=5)

tk.Label(root,text="AmounttoSend").pack()
entry=tk.Entry(root)
entry.pack()

tk.Button(root,text="Send",command=send_money).pack(pady=10)

update_ui()
root.mainloop()
```

OUTPUT:



AftersendingtheETH(3ETH)to‘X’-mywallet remained with 7 ETH